

A-3. Crop Diversification

 Agriculture Land Conservation	 High	 Medium		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions

Measures	Crop diversification is the practice of growing a variety of different crops within a single farming system rather than relying on one single crop. This approach helps farmers reduce risk, improve soil health, and increase overall productivity. It can be achieved by planting multiple crop species in a field simultaneously or through crop rotation, where different crops are grown in sequence over time. Additionally, agroforestry practices and relay cropping, where one crop is planted after another has already started growing are widely practiced crop diversification techniques. Crop diversification is a sustainable farming practice that enhances resilience, improves soil health, and promotes ecological balance. In the CHTs, this practice can provide economic and environmental benefits by reducing risk of crop failure, increase crop yield and improve livelihood of the agriculture dependent communities. Crop diversification also reduces vulnerability of the agriculture sector to various climate change induced hazards.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	Suitable for degraded lands, barren area, smallholder farms, hill slopes								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
Feasibility criteria	- Adequate land space to grow multiple crops and suitable fertility and water retention capacity of the soil. - Choosing crops with similar temperature tolerance and compatible water needs. - Selecting crops based on their pH and nutrients requirements so that different crops can thrive in the same pH condition and not strive for the same soil nutrients.								
Cost	Medium								
Benefits									
	Type	Low		Medium		High			
	Canopy Cover Increase								
	Carbon Sequestration								
	Crop Production Increase								
	Crop Failure Risk Reduction								

	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	- Rotating or intercropping different crops can improve soil fertility and structure, reduce soil erosion, and enhance organic matter due to different crops having varying nutrient requirements, root structures, and growth patterns. - Crop diversification promotes biodiversity, supporting a variety of plants, beneficial insects, and wildlife through aiding pollination and natural pest control.			

	• Crops with complementary water needs optimize water usage, leading to better water conservation and improved moisture retention. • By reducing pest populations through biodiversity, crop diversification lowers the need for chemical pesticides, thus protecting the environment and human health.
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with ensuring food security and creating alternative livelihood opportunities.